United States Patent Application Publication	20250270967
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	UCHIDA; Akito

VEHICLE WITH ENGINE

Abstract

A vehicle, with an engine, includes an engine control unit configured to control the engine in a circuit mode in which a traveling performance of the vehicle is improved, on a basis of a request transmitted from a portable terminal, when the portable terminal operated by a user of the vehicle determines that a current position of the vehicle equipped with the engine is in a circuit, wherein the engine control unit is configured to reject switching to the circuit mode regardless of the request, when the engine control unit determines that a part of combustion conditions among all combustion conditions for stable combustion of fuel supplied to the engine in the circuit mode is not satisfied.

Inventors: **UCHIDA; Akito (Toyota-shi, JP)**

Applicant: **TOYOTA JIDOSHA KABUSHIKI KAISHA** (Toyota-shi, JP)

Family ID: **1000008382958**

Assignee: **TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)**

Appl. No.: **19/001778**

Filed: **December 26, 2024**

Foreign Application Priority Data

JP	2024-026465	Feb. 26, 2024

Publication Classification

Int. Cl.: **F02D41/22** (20060101); **B60K35/10** (20240101); **B60K35/22** (20240101)

U.S. Cl.:

CPC **F02D41/22** (20130101); **B60K35/10** (20240101); **B60K35/22** (20240101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-026465, filed on Feb. 26, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a vehicle with an engine.

BACKGROUND

[0003] A limiter is known to limit various functions for safety setting, fuel saving, and the like of a vehicle. As functions for safety setting, for example, a sideslip prevention function such as an anti-lock brake system is known. As functions for saving fuel, for example, a variable cylinder management system is known in which “one third or half” of the engine is automatically stopped depending on a traveling state to save fuel.

[0004] Further, a technique is known in which a current position of the vehicle is identified from data acquired by a global positioning system (GPS) and map information, and the amount of steering force assist is changed if the vehicle is on a circuit road. In addition, a technique is known in which the above-described limiter is released when it is determined that the vehicle is positioned in a circuit (see, for example, Japanese Unexamined Patent Application Publication No. 2015-199382).

[0005] When the limiter is released in a case where the vehicle is positioned in the circuit, for example, fuel saving is mitigated. However, even if the fuel saving is mitigated, the fuel might not be stably combusted inside the engine depending on, for example, the fuel injection amount, the temperature of intake air taken into the engine, and the like. In this case, even if the mode is switched to the circuit mode for improving the traveling performance of the vehicle on the circuit road, the traveling performance of the vehicle might not be sufficiently improved.

[0006] Therefore, for example, when switching to the circuit mode is permitted, it is desirable to determine in advance whether or not the engine is suitable for all of a large number of combustion conditions for determining whether or not the fuel supplied to the engine is stably combusted. However, if it is individually determined whether all of such combustion conditions are satisfied or not, the number of steps for meeting the combustion conditions might increase. Similarly, even when the switching to the circuit mode is rejected, the number of adaptation steps might increase. If the number of adaptation steps increases, it might take time to adapt the combustion conditions.

SUMMARY

[0007] It is therefore an object of the present disclosure to provide a vehicle, with an engine, reducing the number of adaptation steps for combustion conditions of fuel when switching to the circuit mode is rejected.

[0008] The above object is achieved by a vehicle includes: an engine; and an engine control unit configured to control the engine in a circuit mode in which a traveling performance of the vehicle is improved, on a basis of a request transmitted from a portable terminal, when the portable terminal operated by a user of the vehicle determines that a current position of the vehicle equipped with the engine is in a circuit, wherein the engine control unit is configured to reject switching to the circuit mode regardless of the request, when the engine control unit determines that a part of combustion conditions among all combustion conditions for stable combustion of fuel supplied to the engine in the circuit mode is not satisfied.

[0009] The engine control unit may be configured to turn off the circuit mode, after the vehicle stops so as to stabilize behavior of the vehicle, in a case where the engine control unit determines that the part of the combustion conditions is not satisfied while the engine is controlled in the

circuit mode and the vehicle is traveling.
[0010] The vehicle may further include a display control unit configured to control display of a display device provided in a vehicle cabin of the vehicle, wherein the display control unit may be configured to output a display requesting the vehicle to stop to the display device, on a basis of an instruction transmitted from the portable terminal, when the portable terminal determines that the current position of the vehicle is outside the circuit, and the engine control unit may be configured to turn off the circuit mode, after the vehicle stops so as to stabilize behavior of the vehicle.
[0011] The vehicle may further include a communication control unit provided in the vehicle and configured to control communication between the portable terminal and the engine control unit, wherein the engine control unit may be configured to turn off the circuit mode, when the engine control unit does not receive a signal transmitted from the communication control unit.
[0012] The engine control unit may be configured to exclude determination related to fail-safe control for the engine during a period from a first time to a second time, the first time may be a time when a starter for starting the engine is turned on, and the second time may be a time when a predetermined time elapses from a time when the starter is turned off.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] FIG. **1** illustrates an example of a vehicle control system;
[0014] FIG. **2**A is a view illustrating an example of a correspondence table held by a portable terminal, and FIG. **2**B illustrates an example of a hardware configuration of an engine ECU;
[0015] FIG. **3** is a processing sequence diagram illustrating an example of an operation of the vehicle control system;
[0016] FIG. **4** is a flowchart illustrating an example of a determination process; and
[0017] FIG. **5** is an example of a display device including a stop request display.
DETAILED DESCRIPTION
[0018] Hereinafter, embodiments of the present disclosure will be described with reference to the drawings.
[0019] As illustrated in FIG. **1**, a vehicle control system ST includes a vehicle **10**, a server **20**, and a portable terminal **30**. Although a smartphone is illustrated as an example of the portable terminal **30** in FIG. **1**, a tablet terminal may be used instead of the smartphone. The vehicle **10**, the server **20**, and the portable terminal **30** cooperate with one another, whereby the vehicle control system TS provides a service limited to a circuit C**1** to a driver **10**D who is a user of the vehicle **10**.
[0020] For example, when the vehicle **10** enters the circuit C**1** and the driver **10**D operates the portable terminal **30** in an office C**2** of the circuit C**1** to activate a circuit application, the portable terminal **30** acquires GPS information including a current position of the vehicle **10** via the server **20**. The circuit application is application software installed in the portable terminal **30** and is associated with the vehicle **10**. The circuit application is used when the vehicle **10** is controlled in a circuit mode for improving the traveling performance of the vehicle **10**.
[0021] The vehicle **10** includes a data communication module (DCM) **11** as a wireless communication device to which an antenna ATN is connected, a DCM-electronic control unit (ECU) **12**, and a GPS **13**. The DCM-ECU **12** is an example of a communication control unit, and controls communication between the portable terminal **30** and the engine ECU **15** described later. The GPS **13** measures the positions of the vehicle **10** and holds GPS information including the measured positions. The DCM-ECU **12** acquires the GPS information from the GPS **13** and transmits the GPS information to the server **20** by radio wave WL via the DCM **11** and the antenna ATN. Therefore, when the server **20** requests the vehicle **10** to transmit the GPS information, the server **20** acquires the GPS information from the vehicle **10**.

[0022] The GPS information reaches the server **20** via the base station BS and the communication network NW. The communication network NW includes one or both of the Internet and a local area network (LAN). When the portable terminal **30** requests the server **20** to transmit the GPS information, the server **20** transmits the GPS information to the portable terminal **30** by the radio wave WL via the communication network NW and the base station BS. Thus, the portable terminal **30** acquires the GPS information of the vehicle **10**.
[0023] The server **20** also holds map information (hereinafter referred to as circuit information) including the position or area of the circuit C**1**. When the portable terminal **30** requests the server **20** to transmit the circuit information, the server **20** transmits the circuit information to the portable terminal **30** by the radio wave WL via the communication network NW and the portable base station BS. Thus, the portable terminal **30** acquires the circuit information.
[0024] The portable terminal **30** determines whether the current position of the vehicle **10** is within the circuit C**1** or not on the basis of the circuit information and the GPS information. When the vehicle **10** is not located in the circuit C**1**, the portable terminal **30** rejects the switching to the circuit mode and presents this rejection to the driver **10**D on the screen.
[0025] On the other hand, when the vehicle **10** is located in the circuit C**1**, the portable terminal **30** presents a notice about the switching to the circuit mode to the driver **10**D, and requests the driver **10**D to accept the switching. In this way, in the circuit mode, it is determined whether or not the position of the vehicle **10** is within the circuit C**1**. Therefore, the circuit mode is different from a sport mode (or a sport traveling mode) in which the traveling performance is improved only by switching of a switch provided in the vehicle cabin of the vehicle **10** without performing such determination.
[0026] When the portable terminal **30** receive the acceptance of the switching from the driver **10**D, the portable terminal **30** transmits circuit-mode request information (hereinafter, referred to as request ID (Identifier)) including the acceptance of the switching to the server **20**. The request ID is identification information for requesting the vehicle **10** to switch to the circuit mode. The request ID is prepared and defined for each version of the circuit application. Therefore, when the version of the circuit application is updated by version upgrade, different independent request ID is transmitted based on the update of the version.
[0027] When the request ID is transmitted from the portable terminal **30**, the server **20** generates switching information including the request ID and transmits the switching information to the vehicle **10**. The switching information is information for switching the traveling performance of the vehicle **10** to traveling performance specialized for traveling in the circuit C**1**, which will be described in detail later. For example, the server **20** transmits the switching information to the vehicle **10** by a short message service (SMS).
[0028] In the vehicle **10**, the DCM-ECU **12** receives the switching information from the server **20** via the DCM **11** and the antenna ATN. The vehicle **10** includes an engine **14**, an engine ECU **15**, a display device **16**, and a meter ECU **17**. The engine ECU **15** is an example of an engine control unit. The meter ECU **17** is an example of a display control unit. The engine **14** is provided with a starter **14**A for starting the engine **14**. A speed sensor **15**V for detecting the speed of the vehicle **10** is connected to the engine ECU **15**. The display device **16** is provided in the vehicle cabin of the vehicle **10**. The DCM-ECU **12**, the engine ECU **15**, and the meter ECU **17** achieves the control device for the vehicle **10**.
[0029] When the DCM-ECU **12** receives the switching information, the DCM-ECU **12** transmits the switching information to the engine ECU **15** by using a controller area network (CAN) signal. Thus, the engine ECU **15** receives the switching information. When receiving the switching information, the engine ECU **15** determines whether a first condition and a second condition, which are a part of all of at least three or more conditions for stably burning fuel supplied to the engine **14** in the circuit mode, are not satisfied. The first condition and the second condition are examples of combustion conditions. All of the at least three conditions including the first condition

and the second condition are examples of the combustion condition.
[0030] For example, the at least three conditions include the first condition related to a temperature of coolant that cools the engine **14**, the second condition related to an atmospheric pressure, and a third condition related to an injection amount of the fuel supplied to the engine **14**. The at least three conditions may include, for example, a fourth condition related to a rotational speed of the engine **14**, a fifth condition related to a temperature of the outside air, and a sixth condition related to communication between the DCM-ECU **12** and the engine ECU **15**. The engine ECU **15** determines whether or not, for example, both of the first condition and the second condition are unsatisfied among the first to sixth conditions. In this way, the engine ECU **15** determines whether both the first and second conditions are satisfied or not. Therefore, the engine ECU **15** reduces the number of adaptation steps as compared with the case where it is determined whether or not all the conditions from the first to sixth conditions are satisfied.
[0031] Here, when either one of the first condition and the second condition is satisfied, the engine ECU **15** permits the switching to the circuit mode. Thus, the engine ECU **15** changes the control of the engine **14** based on the request ID included in the switching information. That is, the engine ECU **15** controls the engine **14** in the circuit mode.
[0032] For example, the engine ECU **15** changes a torque upper limit map (hereinafter, simply referred to as torque map) that defines an upper limit of the torque of the engine **14**. Thus, the engine **14** is operated in the circuit mode in which high torque is output. The circuit mode improves the traveling performance of the vehicle **10** as compared to the normal traveling mode. In this way, the use of the circuit mode is accepted, and thus the vehicle control system ST provides a service of conveying the enjoyment of the motor sports to the driver **10**D.
[0033] On the other hand, when both the first condition and the second condition are not satisfied, the engine ECU **15** rejects the switching to the circuit mode. In this case, the engine ECU **15** notifies the portable terminal **30** of the switching error via the DCM-ECU **12**, the server **20**, or the like. Thus, the driver **10**D confirms that the switching to the circuit mode is rejected.
[0034] When the vehicle **10** moves from the inside to the outside of the circuit C**1**, the portable terminal **30** determines that the current position of the vehicle **10** is outside the circuit C**1**. In this case, the portable terminal **30** notifies the server **20** of predetermined information including the fact that the current position of the vehicle **10** is outside the circuit C**1**. The server **20** transmits an instruction including control information for controlling the display of the display device **16** to the meter ECU **17** on the basis of the predetermined information. The meter ECU **17** outputs a display requesting the stopping of the vehicle **10** to the display device **16** on the basis of the instruction transmitted from the server **20**.
[0035] Thus, it is assumed that the driver **10**D shifts to an action of stopping the vehicle **10** by the brake operation. The instruction transmitted from the server **20** reaches the meter ECU **17** via the DCM-ECU **12**, the engine ECU **15**, or the like. The engine ECU **15** determines whether the vehicle **10** is stopped or not on the basis of the speed detected by the speed sensor **15**V. The engine ECU **15** turns off the circuit mode after the vehicle **10** stops. If the engine ECU **15** turns off the circuit mode while the vehicle **10** is traveling, the behavior of the vehicle **10** might not be stable. However, such instability is suppressed, since the engine ECU **15** turns off the circuit mode after the vehicle **10** stops.
[0036] Next, the portable terminal **30** will be described in detail with reference to FIG. **2**A.
[0037] As illustrated in FIG. **2**A, the portable terminal **30** includes a non-volatile memory (NVM) **31**. The NVM **31** stores a correspondence table of version IDs for identifying the versions of the circuit applications and request IDs. For example, a request ID “#**1**” is associated with a version ID “Ver**1**”. A request ID “#**2**” is associated with a version ID “Ver**2**”. Thus, when the version of the application is updated from, for example, “Ver**1**” to “Ver**2**”, the portable terminal **30** transmits the request ID “#**2**”.
[0038] Next, the engine ECU **15** will be described in detail with reference to FIG. **2**B. Hardware

configurations of the DCM-ECU **12** and the meter ECU **17** are basically the same as the hardware configuration of the engine ECU **15**, and thus detailed description thereof will be omitted. The engine ECU **15** indirectly communicates with the portable terminal **30** via the DCM-ECU **12**, the server **20**, or the like.

[0039] The engine ECU **15** is a hardware circuit including a central processing unit (CPU) **15**A, a random access memory (RAM) **15**B, a read only memory (ROM) **15**C, and an input/output interface (I/F) **15**D. The CPU **15**A is an example of a processor and indirectly communicates with the portable terminal **30**. The CPU **15**A, the RAM **15**B, the ROM **15**C, and the input and output I/F **15**D are connected to one another through an internal bus **15**E. Although omitted in FIG. **2**B, the input and output I/F **15**D is connected to the DCM-ECU **12**, the engine **14**, and the meter ECU **17**. At least the CPU **15**A and the RAM **15**B cooperate with each other to realize a computer.

[0040] The software pre-stored in the ROM **15**C is stored in the RAM **15**B mentioned above by the CPU **15**A. The CPU **15**A executes the stored software, and thereby the CPU **15**A executes a series of processes described later. The software may be in accordance with a processing sequence diagram described later.

[0041] The ROM **15**C stores a plurality of torque maps respectively for the request IDs. Each of the torque maps defines the upper limit of the torque of the engine **14**. Since the request IDs are prepared and defined for respective versions of the circuit application, each torque map is stored in the ROM **15**C for each version of the circuit application.

[0042] Next, the operation of the vehicle control system ST will be described with reference to FIG. **3**.

[0043] First, the portable terminal **30** waits until the circuit application is activated (step S**1**: NO). For example, the process waits until the driver **10**D performs a predetermined operation to instruct activation of the circuit application on the icon of the circuit application displayed on the portable terminal **30**. When a predetermined operation is performed on the icon of the circuit application and the circuit application is activated (step S**1**: YES), the portable terminal **30** requests the server **20** and the DCM-ECU **12** for determination information (step S**2**). The determination information is information for determining whether or not the position of the vehicle **10** is within the circuit C**1**.

[0044] For example, the portable terminal **30** directly requests the server **20** to transmit the determination information. On the other hand, the portable terminal **30** indirectly requests the DCM-ECU **12** for the determination information. That is, the portable terminal **30** requests the DCM-ECU **12** for the determination information via the server **20**. When the determination information is requested from the portable terminal **30**, the server **20** transmits the circuit information as the determination information to the portable terminal **30** (step S**3**). When the determination information is requested from the portable terminal **30** via the server **20**, the DCM-ECU **12** transmits the GPS information as the determination information to the portable terminal **30** via the server **20** (step S**4**).

[0045] The portable terminal **30** determines whether or not the current position of the vehicle **10** is within the circuit C**1** (step S**5**). If the current position is not within the circuit C**1** (step S**5**: NO), the portable terminal **30** skips the subsequent processing. In this case, the portable terminal **30** rejects the switching to the circuit mode, and thus the control of the vehicle **10** in the circuit mode is stopped.

[0046] On the other hand, when the current position is within the circuit C**1** (step S**5**: YES), the portable terminal **30** determines whether or not there is an approval for switching to the circuit mode (step S**6**). For example, the portable terminal **30** presents a notice about the switching to the circuit mode to the driver **10**D on the screen of the portable terminal **30**, and requests the driver **10**D to accept the switching. The notes include, for example, an explanation about deterioration of the engine **14**. When the driver **10**D performs an operation to reject the switching to the circuit mode, the portable terminal **30** determines that the driver **10**D does not accept the switching to the circuit mode (step S**6**: NO). In this case, the portable terminal **30** rejects the switching to the circuit mode, and thus the control of the vehicle **10** in the circuit mode is stopped.

[0047] On the other hand, when the driver **10**D performs an operation of accepting the switching to the circuit mode (for example, pressing of a “YES” button as illustrated in FIG. **1**), the portable terminal **30** determines that the acceptance of the switching to the circuit mode is made (step S**6**: YES). In this case, the portable terminal **30** transmits the request ID to the server **20** (step S**7**). More specifically, the portable terminal **30** checks the version ID for identifying the version of the current circuit application installed in the portable terminal **30**, and specifies and transmits the request ID corresponding to the version ID. For example, if the version of the circuit application identified by the version ID “Ver**2**” is installed in the portable terminal **30**, the portable terminal **30** transmits the request ID “#**2**”.

[0048] When the server **20** receives the request ID, the server **20** transmits the switching information to the DCM-ECU **12** (step S**8**). More specifically, when the server **20** receives the request ID, the server **20** generates the switching information including the received request ID and transmits the switching information to the DCM-ECU **12**. When the DCM-ECU **12** receives the switching information, the DCM-ECU **12** transfers the switching information to the engine ECU **15** (step S**9**).

[0049] When the engine ECU **15** receives the switching information, the engine ECU **15** performs a determination process (step S**10**). The determination process is a process of determining whether both of the first condition and the second condition among the at least three conditions are unsatisfied or not. The determination process will be described in detail later. When one of the first condition and the second condition is satisfied, the engine ECU **15** changes the torque map and immediately uses the changed torque map (step S**11**). More specifically, when one of the first condition and the second condition is satisfied, the engine ECU **15** extracts the request ID from the switching information, and specifies and selects the torque map corresponding to the extracted request ID. For example, when the request ID “#**2**” is extracted, the engine ECU **15** specifies and selects one of the plurality of torque maps associated with the request ID “#**2**”.

[0050] If the torque map associated with the request ID “#**1**” is used before the switching information is received, the engine ECU **15** changes the torque map to the torque map associated with the request ID “#**2**”. When the engine ECU **15** changes the torque map, the engine ECU **15** uses the changed torque map for controlling the vehicle **10**.

[0051] The details of the determination process will be described with reference to FIG. **4**.

[0052] As described above, when the engine ECU **15** receives the switching information, the engine ECU **15** determines whether or not the first condition is satisfied (step S**21**). For example, the engine ECU **15** determines whether or not the temperature of the coolant that cools the engine **14** is equal to or higher than a threshold temperature as the first condition. The threshold temperature is set to a coolant temperature at which the fuel supplied to the engine **14** is stably combusted, on the basis of design, experiments, and the like.

[0053] When the first condition is not satisfied because the coolant temperature is lower than the threshold temperature (step S**21**: NO), the engine ECU **15** then determines whether a second condition is satisfied or not (step S**22**). For example, the engine ECU **15** determines whether or not the atmospheric pressure is equal to or higher than a threshold pressure as the second condition. The threshold pressure is set to an air pressure at which the fuel supplied to the engine **14** is stably combusted, on the basis of design, experiments, and the like.

[0054] If the second condition is not satisfied because the atmospheric pressure is smaller than the threshold pressure (step S**22**: NO), the engine ECU **15** rejects the switching to the circuit mode (step S**23**). That is, when both the first condition and the second condition are not satisfied, the engine ECU **15** may not be able to stably burn the fuel supplied to the engine **14**. In such a case, the engine ECU **15** rejects the switching to the circuit mode because the control of the engine **14** in the circuit mode may not be suitable.

[0055] If the switching to the circuit mode is rejected, the engine ECU **15** notifies a switching error (step S**24**), and the determination process is terminated. More specifically, the engine ECU **15** notifies the portable terminal **30** of the switching error. The switching error reaches the portable terminal **30** via the DCM-ECU **12**, the server **20**, and the like. Thus, the driver **10**D confirms that the switching to the circuit mode is rejected. When the switching to the circuit mode is rejected, the engine ECU **15** skips the process of step S**11** described above.

[0056] On the other hand, when either the first condition or the second condition is satisfied (step S**21**: YES, step S**22**: YES), the engine ECU **15** permits the switching to the circuit mode (step S**25**). That is, when the first condition is satisfied because the coolant temperature is equal to or higher than the threshold temperature, the engine ECU **15** permits the switching to the circuit mode. When the second condition is satisfied because the atmospheric pressure is equal to or higher than the threshold pressure, the engine ECU **15** permits the switching to the circuit mode. When the switching to the circuit mode is permitted, the engine ECU **15** executes the processing of the above-described step S**11**, and ends the determination processing.

[0057] In this way, the engine ECU **15** rejects the switching to the circuit mode. when it is determined that both the first condition and the second condition among at least three or more conditions under which the fuel supplied to the engine **14** stably burns are not satisfied. Thus, the engine ECU **15** reduces the number of the adaptation steps, as compared with the case where it is individually determined whether or not all of at least three or more conditions are satisfied. As a result, the conformity time is reduced as compared to the case where it is individually determined whether or not all of at least three or more conditions are satisfied.

[0058] When the engine ECU **15** determines that both the first condition and the second condition are not satisfied while the engine **14** is controlled in the circuit mode and the vehicle **10** is traveling, the engine ECU **15** turns off the circuit mode after the vehicle **10** is stopped. If the engine ECU **15** turns off the circuit mode while the vehicle **10** is traveling, the behavior of the vehicle **10** might not be stable. The engine ECU **15** turns off the circuit mode after the vehicle **10** is stopped, thereby suppressing such instability.

[0059] As described above, when the vehicle **10** move from the inside to the outside of the circuit C**1**, the meter ECU **17** outputs a stop request display **19** for requesting the vehicle **10** to stop to the display device **16** as illustrated in FIG. **5** based on the instruction transmitted from the server **20**. The stop request display **19** is output near a tachometer **18**. Thus, it is assumed that the driver **10**D shifts to an action of stopping the vehicle **10** by the brake operation. The engine ECU **15** turns off the circuit mode after the vehicle **10** stops. If the engine ECU **15** turns off the circuit mode while the vehicle **10** is traveling, the behavior of the vehicle **10** might not be stable. The engine ECU **15** turns off the circuit mode after the vehicle **10** is stopped, thereby suppressing such instability.

[0060] Further, the engine ECU **15** turns off the circuit mode when not receiving the electrical signal transmitted from the DCM-ECU **12**. If the engine ECU **15** does not receive signals from the DCM-ECU **12**, the vehicle **10** may behave differently from the circuit mode specifications. By turning off the circuit mode, the engine ECU **15** avoids such behavior.

[0061] In addition, the engine ECU **15** excludes the determination related to the fail-safe control for the engine **14** from the time when the starter **14**A is turned on to the time after a predetermined offset time elapses from the time when the starter **14**A is turned off. The fail-safe control includes, for example, a limitation on the maximum rotation speed of the engine **14**. The fail-safe control includes a limitation on the amount of air taken into the engine **14**, a limitation on the voltage of a battery mounted on the vehicle **10**, and the like. The offset time is set from several tens of milliseconds to several hundreds of milliseconds.

[0062] The reason why the determination regarding the fail-safe control is excluded is as follows. When the engine **14** is started in the circuit mode while the ignition switch is turned on and the engine **14** is stopped, the circuit mode is released due to a voltage drop that occurs at the time when the starter **14**A is turned on. That is, the circuit mode is dynamically switched from ON to OFF due to the voltage drop. In this case, the driver **10**D is requested to switch the circuit mode to ON again.

As a result, the satisfaction level of the service of conveying the enjoyment of motor sports to the driver **10**D may decrease. By temporarily excluding the determination related to the fail-safe control, the dynamic cancellation of the circuit mode is avoided. This suppresses the above-described decrease in satisfaction.
[0063] Although some embodiments of the present disclosure have been described in detail, the present disclosure is not limited to the specific embodiments but may be varied or changed within the scope of the present disclosure as claimed.
[0064] For example, the first condition and the second condition are used as an example of the specific combustion conditions in the above description. However, the specific combustion condition may be one of the first condition and the second condition. Further, when four or more combustion conditions are employed, the first condition, the second condition, and the third condition may be employed as the specific combustion conditions.

Claims

1. A vehicle with an engine, the vehicle comprising; an engine control unit configured to control the engine in a circuit mode in which a traveling performance of the vehicle is improved, on a basis of a request transmitted from a portable terminal, when the portable terminal operated by a user of the vehicle determines that a current position of the vehicle equipped with the engine is in a circuit, wherein the engine control unit is configured to reject switching to the circuit mode regardless of the request, when the engine control unit determines that a part of combustion conditions among all combustion conditions for stable combustion of fuel supplied to the engine in the circuit mode is not satisfied.
2. The vehicle with the engine according to claim 1, wherein the engine control unit is configured to turn off the circuit mode, after the vehicle stops so as to stabilize behavior of the vehicle, in a case where the engine control unit determines that the part of the combustion conditions is not satisfied while the engine is controlled in the circuit mode and the vehicle is traveling.
3. The vehicle with the engine according to claim 1, further comprising a display control unit configured to control display of a display device provided in a vehicle cabin of the vehicle, wherein the display control unit is configured to output a display requesting the vehicle to stop to the display device, on a basis of an instruction transmitted from the portable terminal, when the portable terminal determines that the current position of the vehicle is outside the circuit, and the engine control unit is configured to turn off the circuit mode, after the vehicle stops so as to stabilize behavior of the vehicle.
4. The vehicle with the engine according to claim 1, further comprising a communication control unit provided in the vehicle and configured to control communication between the portable terminal and the engine control unit, wherein the engine control unit is configured to turn off the circuit mode, when the engine control unit does not receive a signal transmitted from the communication control unit.
5. The vehicle with the engine according to claim 1, wherein the engine control unit is configured to exclude determination related to fail-safe control for the engine during a period from a first time to a second time, the first time is a time when a starter for starting the engine is turned on, and the second time is a time when a predetermined time elapses from a time when the starter is turned off.